MICHAEL KASMAN

Data Scientist

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EDUCATION

B.S. Computer Engineering

The University of Texas at Dallas

August 2018 - Present *Richardson, TX*

Calculus I, II, III • Differential Equations • Linear Algebra • Probability & Statistics • Computer Science I, II • Data Structures & Algorithms

WORK EXPERIENCE

Undergraduate Researcher - Hand Synergies

The University of Texas at Dallas

June 2020 - August 2020 *Richardson, TX*

Achievements/Tasks

- Investigated non-linear dimensionality reduction techniques on hand joint angle measurements for real-time application under Dr. Ann Majewicz Fey & Dr. Edoardo Battaglia
- Prepared hand joints dataset for training and testing on dimensionality reduction models: **principal component analysis (PCA), kernal PCA, UMAP,** and **autoencoder**
- Utilized **Keras, Tensorflow,** and **Sklearn** to build a **stacked autoencoder** with a reconstruction error of **≈ 3.6%** (less than PCA)

Undergraduate Researcher - Robotic Surgery

The University of Texas at Dallas

May 2019 - August 2019 *Richardson, TX*

Achievements/Tasks

- Researched the variance of human-centric metrics between virtual reality simulation and physical dry-lab exercises in robotic surgical training under Dr. Ann Majewicz Fey
- Pre-processed and analyzed data in a Linux environment using **Pandas**, **Seaborn,** and **Pingouin** in **Jupyter** Lab to visualize analysis of variance **(ANOVA)**
- Published work to the Journal of Medical Robotics Research for special issue: Technology-enabled Tools for Clinical Skills Assessment

Clark Foundation Summer Research Scholar

The University of Texas at Dallas

June 2018 - August 2018 *Richardson, TX*

Achievements/Tasks

- Developed a mobile indoors navigation app, Constellation, for the UT Dallas campus by collaborating with undergraduate researchers and principal investigator, Dr. Ravi Prakash
- Created **PHP** scripts to receive HTTP requests from the client and acquire the requested path coordinates and user location from **ArcGIS** and **Cisco CMX**
- Accomplished Constellation 1.0, measured by 50+ successful consecutive tests in navigating a path between rooms, through frontend and backend communication

PUBLICATIONS

Michael Kasman, Ziheng Wang, Marco Martinez, Robert Rege, Herbert Zeh, Daniel Scott, and Ann Majewicz Fey. A Comparative Human-Centric Analysis of Virtual Reality and Dry Lab Training Tasks on the da Vinci Surgical Platform. *Journal of Medical Robotics Research: Special Issue on Technology-enabled Tools for Clinical Skills Assessment.*

SKILLS

Bootstrap | SQL | OpenCV | Flask

LaTex | Linux | NumPy | Matplotlib

Pandas | Sklearn | Keras | Tensorflow

HTML 5/CSS 3	**Proficient**
Python	**Proficient**
JavaScript	**Intermediate**
Java	**Intermediate**
C/C++	**Intermediate**
PHP	**Intermediate**

PROJECTS

Flare

- A React Native mobile application that analyzes the victim density level after a natural disaster from drone/bird-eye viewed images using deep learning

Kaggle Medical Cost Predictor

- A Python Notebook analyzing health insurance data and predicting medical charges from age, sex, BMI, smoker, and the number of children variables

Constellation Mobile Indoors Navigation App

- An Android mobile indoors navigation app capable of location tracking and path-finding between rooms on the UT Dallas campus

GreenView

- An Android mobile app that determines whether an item is recyclable through image recognition with the **Clarifai API**

ACHIEVEMENTS

2020 Jonsson School Undergraduate Research Award

Awarded $500, among five other recipients, for medical robotics research at the University of Texas at Dallas

IEEE-Eta Kappa Nu (IEEE-HKN)

Recognized in the top quarter of class in Computer Engineering and for service in the engineering community at the University of Texas at Dallas

2017 cPanel & FreeCodeCamp Hackathon 1st Place

Won 1st Place with three team members out of 50+ teams for the best mobile app, GreenView, that identifies the recyclability of items

Better World Fund & UNA-USA Thank a Peacekeeper (TAPK) Campaign Award

Awarded a grant for heading the TAPK campaign at the Dallas Area Model United Nations Conference: delivered 300 complete TAPK cards